

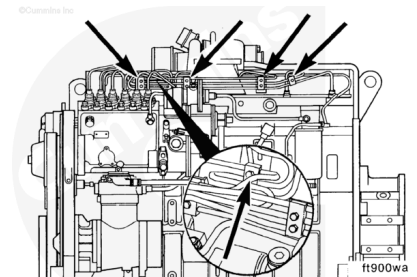
Trainings ▾

General Information

⚠ CAUTION ⚠

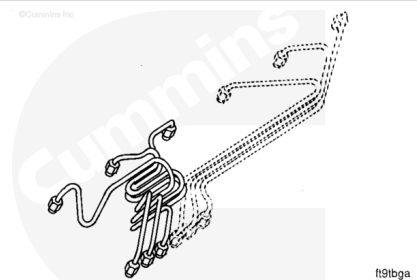
The high-pressure fuel lines will be damaged, unless they are clamped securely and routed so they do not contact each other or any other components.

The high-pressure fuel lines are designed and manufactured to deliver fuel at injection pressure to the injectors. The high-pressure pulses will cause the lines to expand and contract during the injection cycle.



The length, internal size, and rigidity of the high-pressure fuel lines is critical to smooth engine operation. An attached metal tag is used to identify each line with a part number.

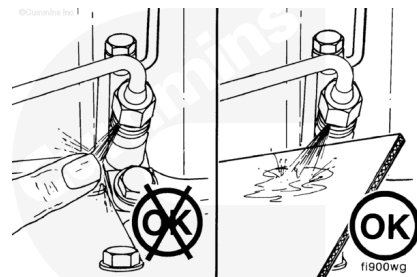
Note : Do not weld or substitute line. Use only the specified part number for the engine.



⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

Use cardboard to check for cracks and leaks in the high-pressure fuel lines. With the engine running, move the cardboard over the fuel lines and look for fuel spray on the cardboard. Fuel leaks can cause poor engine performance.



Preparatory Steps

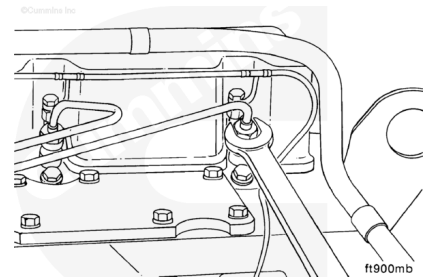
Clean all debris from around the fuel lines.

©Cummins Inc.

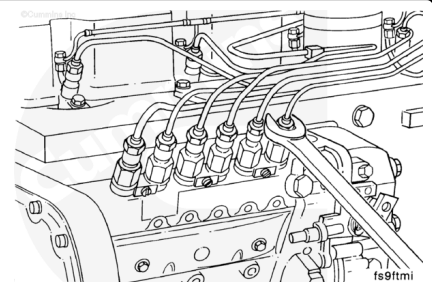
Remove

Note : If individual lines are to be replaced, remove the vibrations isolators from the set of lines containing the line to be replaced.

Disconnect the line(s) from the injector(s).



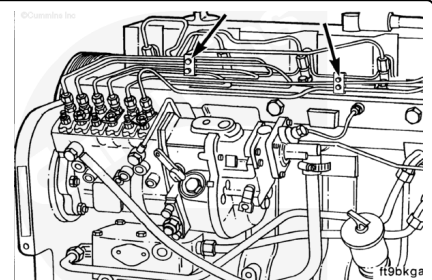
Disconnect the line(s) from the fuel injection pump.



⚠ CAUTION ⚠

The high-pressure fuel lines will be damaged if they are not clamped securely and routed so they do not contact each other or any other component.

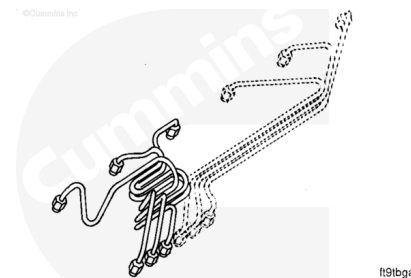
The high-pressure fuel lines are designed and manufactured to deliver fuel at injection pressure to the injectors. The high-pressure pulses cause the lines to expand and contract during the injection cycle.



⚠ CAUTION ⚠

Do not weld or substitute lines. Use only the specified part number for the engine.

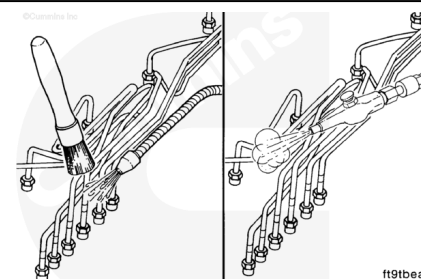
The length, internal size, and rigidity of the lines are critical for smooth engine operation. An attached metal tag is used to identify each line with a part number.



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Wash the fuel lines in clean solvent.

Dry the fuel lines with compressed air.

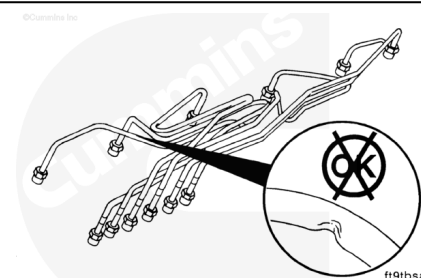
Note : Make sure all paint chips are removed when cleaning the fuel lines.

Inspect for Reuse

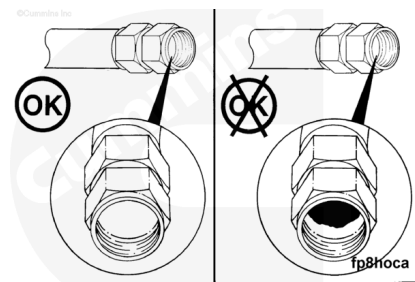
Inspect the high-pressure fuel lines for obvious damage, such as lines that have been bent to facilitate injector removal.

Note : High-pressure pulses expand and contract the injector lines that result in internal flaking at the bent areas.

Bent lines **must** be replaced.

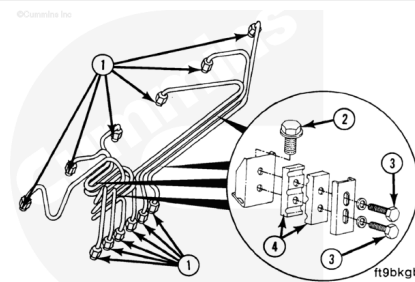


Inspect the high-pressure fuel lines for cracks at both ends.



Inspect all fuel line fittings (1), support bracket capscrews (2), vibration isolator capscrews (3), and isolators (4), for wear and damage.

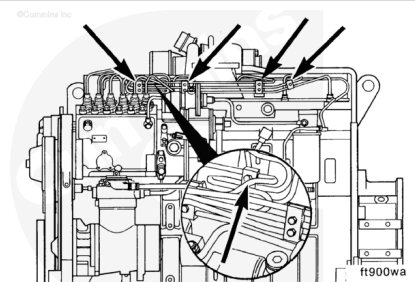
Any damaged parts **must** be replaced.



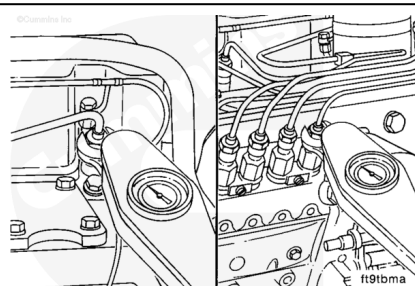
Install

Loosen the vibration isolator capscrews so the fuel lines can be easily moved.

Note : To prevent breakage to the high-pressure fuel lines, they must be connected to the injector and the fuel injection pump in a free state, without forcing the connecting nuts since the fuel lines are properly sized for specific application.

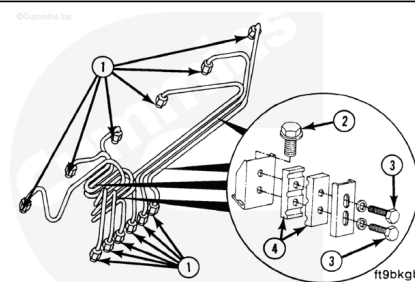


Note : If removed, install the support clamps in their original positions and make sure the high-pressure fuel lines do not contact each other or other components.



Tighten all fittings and mounting hardware.

Fuel line connections	24 n.m	[18 ft-lb]
Support Bracket Capscrews	24 n.m	[18 ft-lb]



Isolator Capscrews	6 n.m	[53 in-lb]
-----------------------	-------	------------

Vent

Bleed the air from the high-pressure fuel lines at the injector connections. Bleed one line at a time until the engine runs smoothly.

Note : If the air can not be removed, check the fuel injection pump and supply line for suction leaks.

